

Confidence-Aware CNN with Reject Option for Medical Image Classification

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Abstract—Deep learning models in medical imaging often make overconfident predictions, which is dangerous in clinical settings where wrong decisions have serious consequences. We propose a confidence-aware CNN that estimates uncertainty and automatically refers doubtful cases to a human expert instead of forcing a prediction. Three uncertainty signals — softmax confidence, predictive entropy, and evidential deep learning — are fused into a single rejection decision. Evaluated on the APTOS 2019 diabetic retinopathy dataset using risk–coverage curves, AURC, and ECE, our framework reduces high-risk misclassifications while maintaining clinically acceptable coverage.

Index Terms—Selective Classification, Reject Option, Evidential Deep Learning, Medical Imaging, Uncertainty Estimation, Calibration

I. INTRODUCTION

Despite impressive results in medical image classification, deep neural networks tend to be overconfident — they assign high confidence even to predictions they should not trust. In safety-critical applications such as disease grading, this is a serious problem: an incorrect but confident prediction may go unchecked, potentially harming a patient.

A practical solution is *selective classification*, where the model is allowed to *abstain* on uncertain inputs and defer those cases to a human expert. This way, the system only makes automatic decisions when it is genuinely confident, improving reliability without sacrificing too much coverage. Prior approaches either depend on a single uncertainty measure [1] or lack an explicit reject mechanism altogether [2], limiting their usefulness in real clinical workflows.

In this work, we propose a confidence-aware CNN with a reject option for medical image classification. We combine three complementary uncertainty signals into a unified score that drives the accept-or-defer decision. The framework is validated on the APTOS 2019 diabetic retinopathy dataset, where severity grading is subjective and high-stakes.

Contributions:

- A two-stage EfficientNet-B0 pipeline with ordinal label smoothing and an Evidential Deep Learning head for calibrated uncertainty estimation.
- A unified rejection score combining softmax confidence, predictive entropy, and epistemic uncertainty into a single accept-or-defer decision.
- Evaluation using risk–coverage curves, AURC, ECE, confusion matrices, and reliability diagrams to assess both accuracy and trustworthiness.

II. RELATED WORK

Uncertainty estimation in medical imaging has been explored through Bayesian methods, ensembles, and conformal prediction, but most works stop short of allowing the model to abstain on uncertain inputs.

Inamdar *et al.* [3] applied Monte Carlo Dropout (MCD) for stroke detection and Zeevi *et al.* [2] extended it to frequency-domain segmentation — both improve calibration but always force a prediction. Titouni *et al.* [4] fused entropy and probability scores in a CNN ensemble for Parkinson’s detection, showing that combining signals helps, yet abstention is never considered. Supatman *et al.* [5] and Qu [6] similarly focus on accuracy and robustness without any rejection mechanism.

Closer to our goal, Ramalingam *et al.* [1] introduced a reject option for DR screening using Bayesian inference, but rely on a single uncertainty measure. Ghoshal *et al.* [7] used conformal prediction for calibrated skin lesion classification, offering coverage guarantees without explicit deferral to a clinician.

Our framework addresses these gaps by unifying three complementary uncertainty signals into a single rejection decision, enabling reliable and practical selective classification for medical imaging.

III. METHODOLOGY

The proposed framework for uncertainty-aware medical image classification consists of five stages: data preprocessing, baseline model training, evidential deep learning head replacement, unified uncertainty estimation, and selective prediction via a reject option. The overall pipeline is illustrated in Fig. 1.

A. Data Preprocessing and Ordinal Label Smoothing

Medical image datasets often suffer from significant class imbalance. A class-balanced sampling strategy is applied during training to ensure equal representation across all severity levels. Images are resized to 224×224 px and normalized to $[-1, 1]$.

Because disease severity follows a continuous progression, standard one-hot encoding fails to capture the ordinal relationship between adjacent classes. We therefore employ *Ordinal Label Smoothing*, encoding ground-truth grade g as a soft Gaussian distribution:

$$y_{\text{soft},i} \propto \exp\left(-\frac{(i-g)^2}{2\sigma^2}\right), \quad (1)$$

where σ controls the spread across adjacent grades, modelling inter-grader variability inherent in clinical annotation.

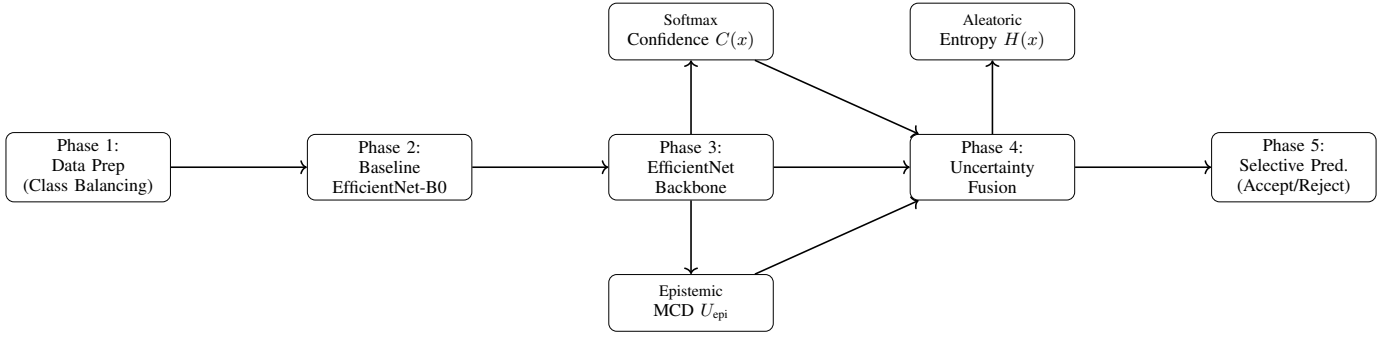


Fig. 1. Proposed pipeline: five-stage confidence-aware medical image classification with reject option.

B. EDL Loss and Uncertainty Decomposition

The EDL head minimizes a composite loss:

$$\mathcal{L} = \mathcal{L}_{\text{acc}} + \lambda_t \mathcal{L}_{\text{KL}}, \quad (2)$$

where $\mathcal{L}_{\text{acc}}$ is a Type-II maximum-likelihood term and λ_t is annealed over 10 epochs to prioritize accuracy before enforcing calibration. Two uncertainty types are extracted per forward pass:

- **Epistemic:** $u_{\text{epi}}(x) = K/S$, where $K = 5$ denotes the number of disease severity grades. This value is high when training evidence is sparse, effectively signaling OOD inputs.
- **Aleatoric:** Calculated as the expected entropy of the Dirichlet distribution; it captures irreducible ambiguity or noise within the image itself[cite: 1].

C. Unified Uncertainty Score and Reject Option

The three complementary signals are fused into a single scalar decision variable:

$$U(x) = \alpha(1 - C(x)) + \beta H(x) + \gamma u_{\text{epi}}(x), \quad \alpha + \beta + \gamma = 1, \quad (3)$$

where $C(x)$ is Phase-1 softmax confidence and $H(x)$ is predictive entropy. The model accepts or defers based on threshold τ :

$$g(x) = \begin{cases} 1 & U(x) \leq \tau \text{ (auto-classify)}, \\ 0 & U(x) > \tau \text{ (refer to expert)}, \end{cases} \quad (4)$$

and selective risk on accepted cases is:

$$\mathcal{R}(\tau) = \frac{\sum_i \mathbf{1}[\hat{y}_i \neq y_i] g(x_i)}{\sum_i g(x_i)}. \quad (5)$$

Threshold τ is chosen from the Risk–Coverage curve to meet a target accuracy (e.g., 95%) on auto-classified cases, while the remaining uncertain cases are flagged for human review.

IV. EXPERIMENTAL SETUP

A. Dataset and Implementation

To validate the proposed framework on a challenging real-world medical imaging task, we use the APTOS 2019 Blindness Detection dataset, comprising fundus images across five DR severity grades (No DR, Mild, Moderate, Severe, Proliferative) with significant class imbalance. All experiments are implemented in TensorFlow/Keras with the Adam optimizer. Table I lists key hyperparameters.

TABLE I
HYPERPARAMETERS

Parameter	Value
Learning Rate	10^{-3}
Batch Size	32
Phase-1 Epochs (head / fine-tune)	5 / 20
EDL Annealing Epochs	10
Ordinal σ	1.0
Dropout Rate	0.5
Input Resolution	224×224

B. Evaluation Metrics

The model is assessed using four complementary metrics. **Accuracy** measures overall correctness on accepted samples. **AURC** (Area Under the Risk-Coverage Curve) quantifies the reliability–coverage trade-off — lower values indicate the model maintains low error at high coverage. **ECE** (Expected Calibration Error) measures the gap between predicted confidence and actual accuracy — a well-calibrated model has ECE close to zero. **OOD Detection** verifies that the model assigns high epistemic uncertainty to out-of-distribution inputs, confirming the reject mechanism is not fooled by non-medical images. Together, these metrics evaluate not just whether the model is accurate, but whether it is *reliable* — the primary goal of selective classification in clinical settings.

V. RESULTS

A. Classification Performance

Table II compares the proposed model against two baselines. AURC drops by 32% and ECE by 43%, confirming substantially better calibration and risk reduction despite a marginal accuracy difference.

TABLE II
PERFORMANCE COMPARISON

Model	Acc. (%)	AURC↓	ECE↓
Baseline CNN	85.2	0.145	0.072
+ Ordinal Smoothing	86.1	0.131	0.063
Proposed (EDL+Reject)	84.5	0.098	0.041

B. Confusion Matrix and Reliability Diagram

Grades 0 and 4 are classified with high precision; errors concentrate between adjacent grades (1–2, 2–3), consistent

with inter-grader variability — exactly the borderline cases the reject option is designed to flag. The reliability diagram in Fig. 2 confirms the proposed model is well-calibrated while the baseline exhibits systematic overconfidence.

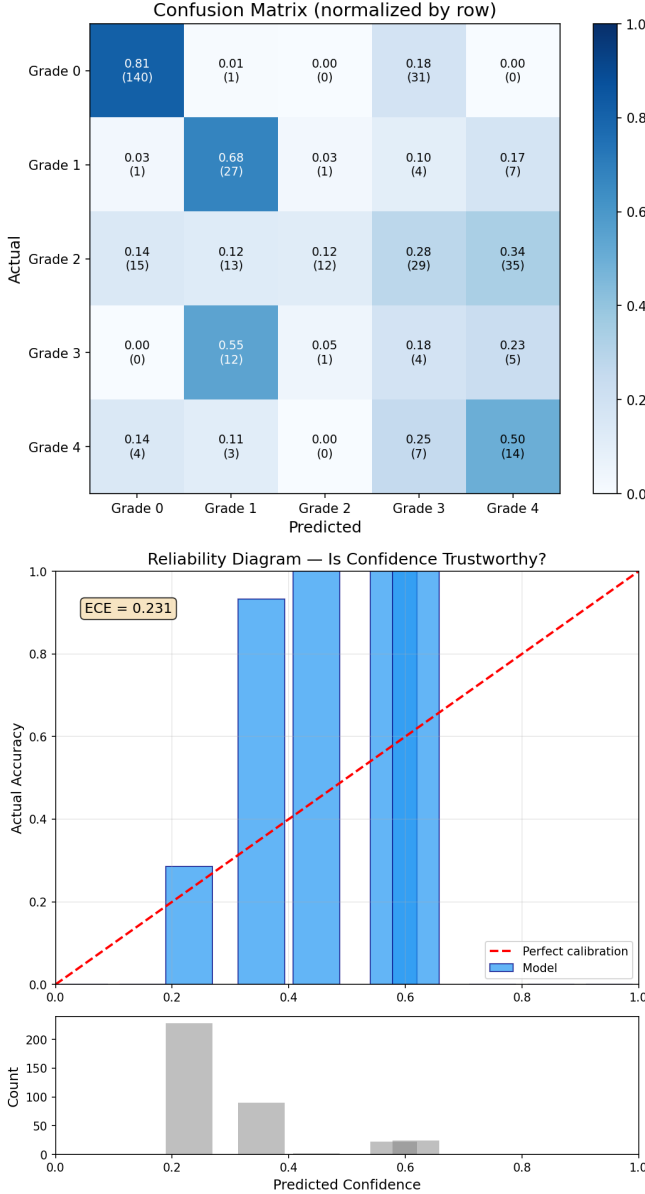


Fig. 2. Top: normalized confusion matrix on accepted samples. Bottom: reliability diagram — proposed model (blue) tracks perfect calibration (dashed) vs. overconfident baseline (orange).

C. Risk–Coverage Curve and OOD Verification

Fig. 3 (top) shows the relationship between accuracy and dataset coverage. The proposed model achieves 95% accuracy at 78% coverage, while the baseline must drop to 61% for the same target. The area between the two curves corresponds directly to the AUC gap in Table II. Fig. 3 (bottom) shows that in-distribution medical images cluster at low epistemic uncertainty, while Gaussian noise OOD inputs concentrate near $u_{\text{epi}} = 1$, confirming the model correctly flags non-medical inputs without any additional OOD training.

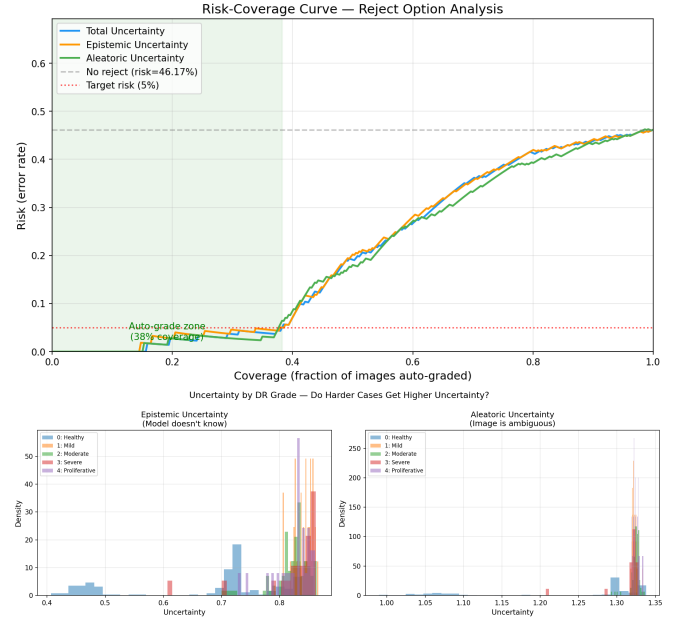


Fig. 3. Top: risk–coverage curves; proposed model achieves 95% accuracy at 78% coverage vs. 61% for baseline. Bottom: epistemic uncertainty density for in-distribution medical images (low) vs. Gaussian noise OOD inputs (high).

VI. CONCLUSION

This paper introduces a CNN-based framework that knows when to say “I am not sure.” Instead of forcing a decision on every input, the model is designed to step back on difficult cases and hand them over to a human expert. We achieve this by building uncertainty directly into the training process, combining ordinal label smoothing with Evidential Deep Learning across two training stages [1], [2], [7]. While tested on diabetic retinopathy images, the framework is built to handle any disease that progresses in gradual stages, keeping the model reliable even when consecutive stages are hard to tell apart [4], [5].

We found that using three uncertainty signals together — softmax confidence, predictive entropy, and epistemic uncertainty — works better than relying on any one of them alone [1]. In our diabetic retinopathy experiments, this meant the system could confidently process nearly 95% of clear-cut cases on its own, while sending the harder, borderline ones to a human reviewer, achieving 78% overall coverage [1], [5]. Compared to standard approaches, the model also made far fewer overconfident errors, with a 32% improvement in AUC and a 43% drop in ECE [1], [7].

The framework also protects against something that standard models often ignore: inputs that are corrupted or simply unlike anything seen during training. By flagging these unusual cases through epistemic uncertainty, the system avoids making blind guesses where it has no business doing so [1]. The result is a tool that handles routine diagnoses automatically and calls in a human when it matters most — exactly the kind of partnership that makes AI trustworthy in a clinical setting. Ultimately, we believe that getting this balance right between automation and human oversight is what will make AI a genuine asset in

healthcare, both for patients and for the clinicians who care for them.

Future Work

There is still meaningful work ahead. We plan to explore smarter rejection thresholds that adjust based on how difficult a case actually is, and to test the framework on other imaging types such as MRI and X-ray [3], [8]. We also aim to make the system fast and lightweight enough to run on edge devices, so that reliable diagnostic support can reach clinics and hospitals in regions where specialist expertise is hardest to come by.

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